Student Housing Rent Price Tiers Classification Model

Ethan Marshall, Salome Collante-Ramirez, and Andrés Ramírez

CSC 642: Statistical Learning with Applications

Professor Vanessa Aguiar-Pulido

April 14th, 2026

Table of Contents

Introduction.....3
State-of-the-Art.....4
Materials and Methods.....5
Datasets.....5
Methods.....7
Evaluation.....7
Concept Maps.....8
Results.....9
Conclusion.....18
References.....19

Introduction

Housing is one of the most unique and principal concerns for the modern college student. Not only is it a major consideration for new freshmen choosing a school, but existing students and graduate students often contend with housing issues on a yearly basis. Beyond this, cities in general are becoming more and more expensive, especially post-COVID. Even in suburban areas, rent in adequate buildings continues to climb as large corporate developments become commonplace. In general, this has led to a general complication in the apartment hunting process, particularly for university students who must contend with roommate-finding, already tight budgets, loans, and dorm requirements.

Especially at the University of Miami, which features dorms in addition to a wide budgetary range of off-campus housing, options can quickly become overwhelming for already busy students. In our project, we hope to use statistical methods to illuminate this chaotic market, especially for international, out-of-state, and prospective students who are especially unfamiliar with the Miami housing world. In this way, we can support these students in finding housing that more closely matches their budgets and amenity "wish lists", even from afar or without domain knowledge of the Miami/Coral Gables area.

Since even new students are likely to understand their budgetary requirements and desired amenities (close to campus, parking, pool, etc.), we can use our methods to classify housing price ranges to fit budgets, as well as to predict property features based on a given student's budget range. In this way, we can help even unfamiliar students make more informed and less rushed housing decisions.

These goals led us to our main research questions. Firstly, as a prerequisite to further analysis, **can we successfully classify housing price ranges into distinct groups such that they represent UM students' budgets?** Then, given that information, **can we predict student housing costs based on desired location, property features, and proximity to UM, all in context of the university housing "bubble"?** A successful application of statistical methods to these questions could yield helpful and relevant results to many members of the UM community.

State-of-the-Art

Real estate econometrics has been a long-studied problem, and one that has historically suffered significant limitations due to the unique and chaotic patterns typical in property markets, especially in isolated "bubbles" like university areas. The desire to be close to a specific location, as well as "ultra" luxury options can highly skew prices out of proportion to their actual features (like square footage) and make generalized regression-based predictions unreliable at best or impossible at worst.

Recent attempts to attack this problem in new ways have made it clear that classification is the key to avoiding these limitations. By grouping general products into tiers based on price (like "cheap", "standard", and "luxury") our model can better reflect consumers' mental models (Gao et al., 2026). And when it comes to properties specifically, a shift to classification yields massive accuracy improvements (Alshammari, 2023).

Beyond general real estate, it has also been shown that highly localized student markets require special consideration. Specifically, by utilizing random forest methods, high accuracy was achieved compared to more traditional methods (Abuh, Ochepa, & Rufai, 2024). Jaroensittichai et al. (2024) showed similar results for an off-campus student "bubble", that classification is an ideal choice for complex, isolated markets like student housing which operate under unique preference and seasonal patterns.

Not only this, but the US Census Bureau (2023) showed that out of several methods tested (logistic regression, KNN, random forest, support vector machines, and gradient boosting), a random forest classifier model performed the best, featuring 83% accuracy, 81% precision, and 87% recall scores. While this working paper focused on gentrification, its conclusions inform problems such as ours.

Materials and Methods

Datasets

To accurately model a relationship between budgets and real estate, this project uses two datasets. Our first represents the data for housing price ranges and the other represents the data for the student budgets in Florida.

Before any analysis though, we noted that there is a significant dataset limitation at play due to price spikes which have occurred post-COVID. For instance, our “apartments for rent” dataset as described below is from 2019, and thus may not accurately represent the 2026 landscape of Miami living.

The first dataset, sourced from the UCI repository (Apartment for Rent Classified, 2019), originally contained 98,924 rows and required extensive preprocessing before analysis. After filtering for Florida-only data points and narrowing the cities between West Palm Beach and Key West, 771 rows remained across 22 columns. Numerical variables: Bathrooms, Bedrooms, Square_Feet, Longitude, and Latitude were set to appropriate int/float types, while the Amenities column, originally a list of strings, was converted to an integer count (see Table 1). Two columns were then engineered: First Distance_to_UM, calculated using R terra library to add relevance for University of Miami students.

Second Value_Category, derived from the value_score which is the summing of normalized values of Bathrooms, Bedrooms, Square_Feet, and Amenities minus normalized price.

$$Value_Score = \sum[norm(variables) \cdot weight] - norm(price)$$

Then the scores are partitioned into three categories, Cheap for affordable/good deals, Standard for normal, and Luxury for expensive/bad deals. This logic follows the "what you get vs. what you pay" logic. Following the removal of two null-value rows and irrelevant columns, the final dataset comprised 769 rows and 8 columns.

Table 1
Dataset for Model A

Variables	Description
Value_Category	Categorical variable
Bathrooms	Int
Bedrooms	Int
Latitude	Float
Longitude	Float
Amenities	Int
Square_Feet	Int
Distance to UM campus	Float

Note. Model A variables after preprocessing

As for our second dataset, the raw student budget data from IPUMS USA (Ruggles et al., 2024) came as two files: a *.dat* file and an *.xml* file. This pre-processing stage served not only to clean the data but also to create a *.csv* file for enhanced functionality. Several columns were removed, and others were kept just in case they are useful in the future. Specifically, filters for Florida and age were set, as well as computed columns to allow for linkage with our first dataset.

For this dataset, we set columns to numeric values, then again filtered for Florida. We also set grade level (undergraduate vs. graduate), current rent, county, location name, and campus distance as columns to keep, while filtering for ages 18-30. This resulted in reducing 22 columns to 11 columns, and 39,801 rows to 1,219 rows.

Out of these 11 columns in the post-processing database, only 4 were used for modeling. The goal of the models is to classify students into budget tiers based on what they pay in rent, so variables that did not directly relate to housing cost or student demographics were left out.

Finally, the “price category” computed column was added by filtering the “price” column given our defined ranges.

Table 2
Breakdown of Price_Category

Price_Category	Price_Range $
Cheap	0 - 1600
Standard	1601 - 2000
Luxury	2001+

Note. Quantiles of price variables distribution for model B

Methods

Based on our initial analysis and the current state-of-the-art, we framed our research questions as a multi-class classification problem. In doing so, we categorized data into our established “cheap”, “standard”, and “luxury” budget tiers.

However, to resolve our two distinct datasets, our categorical variables were used as a unifier to connect the disparate data from each. Applying this to both the original housing inventory data as well as the student demographic data, we were able to classify all instances into our defined tiers. With this, we can correlate student finances with ground data from the housing market under our unified “standard” of price tiers.

As for our procedure, we applied three methods for classifying the data into our tiers: a linear approach with Linear Discriminant Analysis (LDA), a distance based approach with K-Nearest Neighbors (KNN), and a tree based approach with Random Forest.

Evaluation

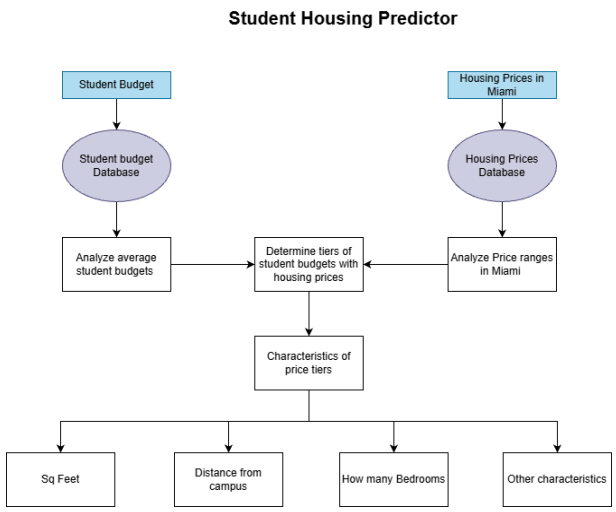
To evaluate our chosen methods, we employed a hold-out testing 80/20 approach as well as 10-fold cross validation. For classification, we also utilized confusion matrices, ROC curves, and AUC scores for a more visual understanding. As ROC is typically a binary visualization, a

one-versus-all approach was utilized to evaluate our multi-class budget tiers. Confusion matrices allowed us to identify specific “troublesome” tiers for the models, while test accuracy provided a more general metric.

Concept Maps

Figure 1

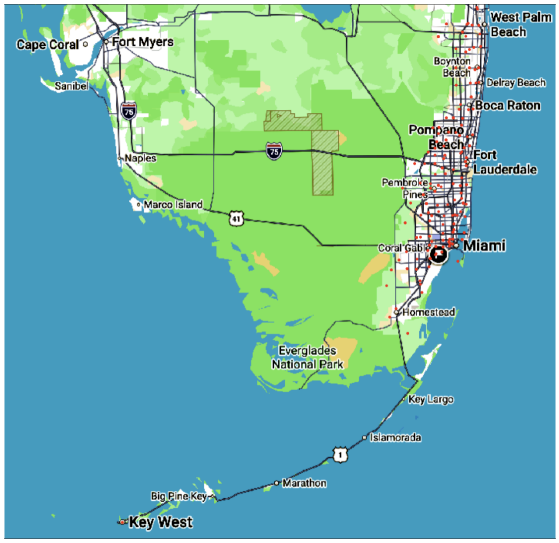
Workflow Diagram



Note. The combination of datasets and features we used to produce our final outputs.

Figure 2

Data Projected Over South Florida Map



Note. Geographic distribution of our dataset across South Florida

Results

Student Budget Analysis (Model B)

First and foremost, our analysis of the reported rent dataset revealed a median rent of $1,800 for south Florida students. Interestingly, we observed generally higher rents for Palm Beach and Miami-Dade counties compared to Broward County. Further, comparing "grade levels" (undergraduate vs. graduate) revealed very similar rent burdens.

From this, we categorized budgets into our three categories: "cheap" (490 students, $0-$1600), "standard" (288 students, $1,601-$2,000), and "luxury" (441 students, $2,001+). In validating our tiers, we performed KNN, LDA, and random forest methods using 11-fold cross-validation. Surprisingly, all of our models performed well but our random forest implementation obtained a strong CV accuracy of 100%. In many classification problems, this can actually indicate overfitting, but since our "price category" unification variable was derived directly from rent data, our model simply learned the selected cutoffs.

Table 3
Model Parameters

Models	Parameters
LDA	Standard LDA, no interaction terms or transformations
KNN	Scale = True - distance sensitive K = 1, K = 11 K = 5 (CV)
RF	Ntree = 500 Importance = True Mtry = default (√p)

Note. Models were implemented following standard classification procedures with parameters selected to balance accuracy and generalizability

Figure 3
Linear Discriminant Analysis Model Confusion Matrix

	Cheap	Standard	Luxury
Cheap	87	4	0
Standard	5	64	16
Luxury	0	0	68

Note. LDA perfectly classified the Luxury tier with zero misclassifications, with the highest misclassification occurring in the Standard tier, where 21 observations were incorrectly assigned.

Figure 4
K-Nearest Neighbors (k=1) Model Confusion Matrix

	Cheap	Standard	Luxury
Cheap	92	4	0
Standard	0	64	6
Luxury	0	0	78

Note. KNN k=1 achieved perfect separation of the Luxury tier with zero misclassifications, with minor errors occurring in the Cheap and Standard tiers.

Figure 5
K-Nearest Neighbors (k=11) Model Confusion Matrix

	Cheap	Standard	Luxury
Cheap	92	2	0
Standard	0	66	5
Luxury	0	0	79

Note. KNN k=11 produced comparable results to k=1, with marginal improvements across all tiers, reducing misclassification in both the Cheap and Standard tiers while achieving perfect separation in the Luxury tier.

Figure 6
Random Forest Model Confusion Matrix

	Cheap	Standard	Luxury
Cheap	92	0	0
Standard	0	68	0
Luxury	0	0	84

Note. Random Forest achieved perfect classification across all three budget tiers with zero misclassifications.

Table 4

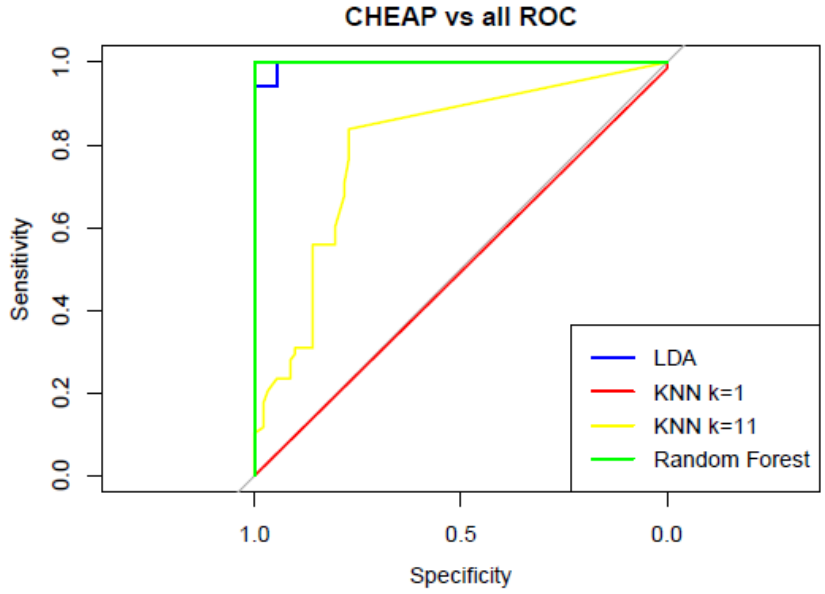
Model B Evaluation Results

Model	Train Accuracy	Test Accuracy	10 Fold CV	AUC
LDA	0.9241	0.8975	0.9224	0.9918
KNN (k=1)	1.0000	0.9590	0.9428	0.5000
KNN (k=11)	0.9672	0.9713	0.9428	0.6906
RF	1.0000	1.0000	1.0000	1.0000

Note. KNN AUC scores are limited due to knn() only returning the winning class vote proportion rather than full class probabilities. Test accuracy is the primary metric for evaluating KNN performance.

Figure 7

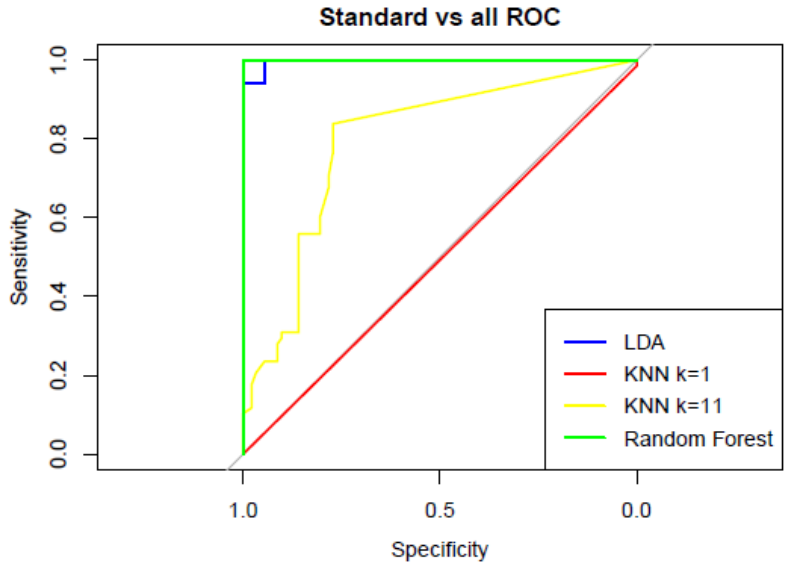
AUC ROC (Model B) for Cheap Class vs All



Note. The Cheap tier shows the clearest separation, with LDA and Random Forest both achieving near-perfect discrimination.

Figure 8

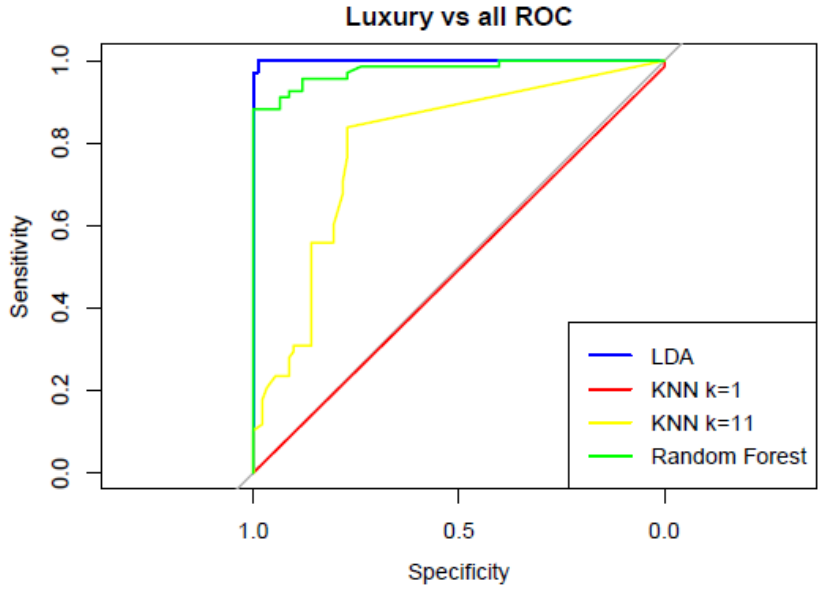
AUC ROC (Model B) for Standard Class vs All



Note. The Standard tier proved the most challenging to discriminate against, yet LDA and Random Forest still maintained strong performance given its narrow $400 price range.

Figure 9

AUC ROC (Model B) for Luxury Class vs All



Note. All models except KNN k=1 show improved discrimination for the Luxury tier compared to Standard, reflecting the open-ended nature of the Luxury range above $2,001.

Thus, with this high accuracy, we are able to conclude that our "price category" is an ideal unifier for our data, given its non-overlapping ranges and its consistency within our original dataset.

Housing Market Classification (Model A)

After validating our budget categories and the value_score for this model, we applied the tiers to the processed Florida housing data. To better visualize the reach of our model, we mapped the apartments in relation to the UM campus, which is labeled with a flag token (see Figure 2). Based on physical features and proximity, we hoped to classify properties into their correct tiers. In this case, we used a hold-out based test set for evaluation. Also, each algorithm was tuned to perform better than default settings (see Table 5).

As before, and as in the current state-of-the-art, random forest emerged as superior, achieving a test accuracy of about 80%. According to our confusion matrix, while the model experienced some overlap between borderline standard/cheap and standard/luxury properties, it successfully avoided extreme misclassifications. LDA yielded 66%, featuring more severe misclassifications.

Table 5
Model Parameters

Models	Parameters
LDA	Interaction terms: square_feet*longitude, square_feet*latitude Non-linear transformation: log(square_feet) Method = "mve" - robust for outliers
KNN	Scale = True - distance sensitive Kernel = "gaussian" K values
RF	Variable importance = True Ntree = 300 Mtry = sqrt(7)

Note. Default parameters for each model were tuned for a more accurate generalizability.

Figure 10

Linear Discriminant Analysis Model Confusion Matrix

	Cheap	Standard	Luxury
Cheap	24	6	1
Standard	23	46	19
Luxury	3	0	30

Note. LDA has minimal misclassification for Cheap and Luxury class. However, it struggles with the Standard class as it significantly overlaps with Cheap.

Figure 11

K-Nearest Neighbors (k=1) Model Confusion Matrix

	Cheap	Standard	Luxury
Cheap	39	10	0
Standard	9	34	7
Luxury	2	8	43

Note. KNN with k=1 is a base model and shows some misclassification all across the classes.

Figure 12

K-Nearest Neighbors (k=11) Model Confusion Matrix

	Cheap	Standard	Luxury
Cheap	34	8	1
Standard	12	39	5
Luxury	4	5	44

Note. Adjusted k=11 neighbours proves better than the base model in Cheap and Luxury class.

Figure 13
Random Forest Model Confusion Matrix

	Cheap	Standard	Luxury
Cheap	38	7	1
Standard	10	38	4
Luxury	2	7	45

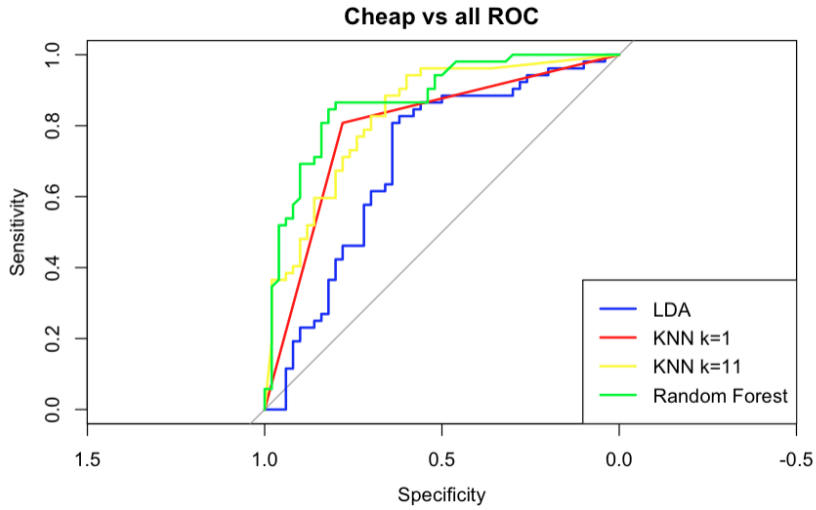
Note. RF shows the least misclassification performance.

Table 6
Model A Evaluation Results

Model	Train Accuracy	Test Accuracy	10 Fold CV	AUC
LDA	0.6548	0.6579	0.7568	0.8366
KNN (k=1)	0.9984	0.7632	0.8058	0.8235
KNN (k=11)	0.8752	0.7697	0.8058	0.8983
RF	0.9968	0.7961	0.8273	0.9191

Note. RF did the best, k=11 is the second best, k=1 is overfitting, and LDA has decent performance.

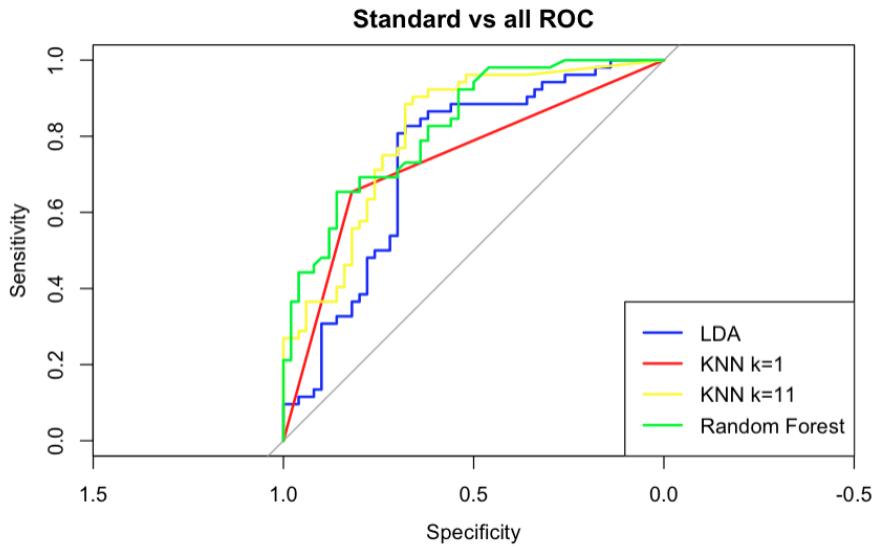
Figure 14
AUC ROC (Model A) for Cheap Class vs All



Note. RF has the best discriminator for Cheap class and LDA struggles the most.

Figure 15

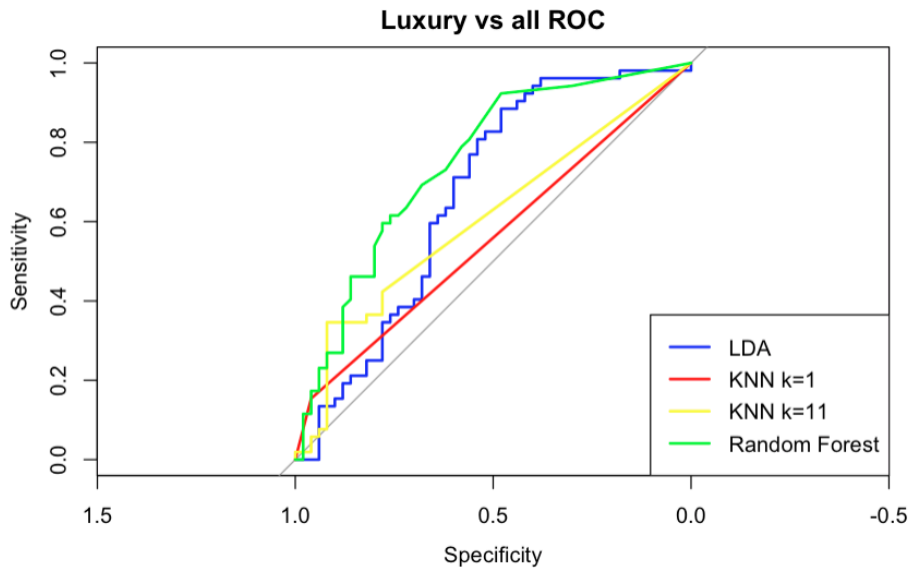
AUC ROC (Model A) for Standard Class vs All



Note. All models have similar performance, RF does show a better elbow shape.

Figure 16

AUC ROC (Model A) for Luxury Class vs All



Note. KNN falls apart and Random Forest keeps the best performance.

The comparison of test results is shown on Table 6, and from it we can prove our model performance. The LDA results overall indicate solid generalization with low performance, as it struggles to distinguish the Cheap class (see Figure 14). As for our KNN results, we observed

severe overfitting and memorization of training data with k = 1. Using k = 11 yielded better results for AUC score and the heavy reduction of overfitting. Additionally, both KNN models suffered with the luxury class (see Figure 16) due to the distance based model being negatively impacted by the lack of normalization.

Our RF had the highest results for all tests, indicating strong class discrimination, as well as higher accuracy but with some overfitting.

As for the ROC results, as ROC is typically a binary visualization, a one vs. all approach was used in generating our versions.

On a macro-level, our results align with current literature, confirming the random forest methods are the most capable when dealing with the complex feature interactions typical in “bubble” markets like the niche world of University of Miami housing.

Conclusion

From a broad standpoint, our random forest model achieved the best results amongst our explored methods, able to handle the complex Miami market features and “bubbles”. This is consistent with prior work in the field, validating our work as a generally accepted approach. Further, classification proved by far the most robust for handling wildly varying housing datasets, another finding consistent with literature.

Interestingly, our “price category” and “value category” variables were successfully able to unify our two separate datasets, which allowed for single analysis of student demographics and ground-truth listings as one.

Continuing work on this project could include up-to-date data (especially featuring post-COVID information), a more highly-tuned amenities variable, and normalized KNN data, all of which could improve results and make developed tools more robust. We specifically envision a developed mobile or web app which would use our methods to provide robust property filters, allowing anyone to find properties which meet their needs, find like-minded roommates, or discover “deals” in certain property types.

References

Abuh, S. O., Ochepa, F. O., & Rufai, M. A. (2024). Enhancing Rental Cost Predictions for Student Housing in Lokoja: A Comparative Analysis of Machine Learning Models. *Asian Journal of Engineering and Applied Technology*, 13(1), 37–43. Retrieved April 12, 2026, from https://doi.org/10.70112/ajeat-2024.13.1.4240

Alshammari, T. (2023). Evaluating machine learning algorithms for predicting house prices in Saudi Arabia. *2023 International Conference on Smart Computing and Application (ICSCA)*, 1-5. Retrieved April 12, 2026, from https://www.researchgate.net/publication/369849167_Evaluating_machine_learning_algorithms_for_predicting_house_prices_in_Saudi_Arabia

Apartment for Rent Classified [Dataset]. (2019). *UCI Machine Learning Repository*. Retrieved April 12, 2026, from https://doi.org/10.24432/C5X623

Gao, R., Xiao, F., Li, J., & Cui, S. (2026). Textual Semantics and Machine Learning Methods for Data Product Pricing. *The Review of Mathematical Economics*, 1(1), 129–167. Retrieved April 12, 2026, from https://doi.org/10.4310/rme.260221154937

Jaroensittichai, K., Jittrong, K., Arreeras, T., Singkhala, E., & Vuttipittayamongkol, P. (2024). Predictive Modeling of Off-Campus Student Housing Rentals Using Machine Learning. *2024 International Conference on Decision Aid Sciences and Applications (DASA)*, 1-5. Retrieved April 12, 2026, from https://doi.org/10.1109/DASA63652.2024.10836318

Ruggles, S., Flood, S., Sobek, M., et al. (2024). *IPUMS USA: Version 15.0* [Dataset]. Minneapolis, MN: IPUMS. Retrieved April 12, 2026, from https://doi.org/10.18128/D010.V15.0

United States Census Bureau. (2023). *Identifying Gentrification Using Machine Learning* (Working Paper SEHSD-WP-2023-15). Retrieved April 12, 2026, from https://www.census.gov/library/working-papers/2023/demo/SEHSD-WP-2023-15.html